Arduino & External LEDs

Learn & Light Up!

A Fun Electronics Lesson for Grade 5

■ Discover how Arduino, LEDs, and resistors work together
■ Learn to build your very first light-up circuit
■ Test your knowledge with a 20-question quiz at the end!

Name: ________________________________

Class: ________________________________

■ Lesson: Getting to Know Arduino & LEDs

■ What is Arduino?

Arduino is a tiny computer that helps us control lights, motors, and more! It lets us make cool projects with electronics and coding.

■ What is an LED?

LED stands for **Light Emitting Diode**. It is a special light that glows when electricity passes through it.

■ Why Use External LEDs?

- Brighter light
- Different colors available
- Can be placed anywhere in a project
- More fun to build circuits!

■ What is a Resistor?

A resistor controls the amount of electricity flowing in a circuit. It keeps our LED safe!

■ Why Do We Need Resistors?

Without Resistor	With Resistor
LED might burn out! Too much electricity can damage it.	LED stays safe and glows just right! Perfect brightness.

■ How to Calculate Resistor Value

R = (V source – V LED) / I LED

V source = 5V (Arduino voltage)

V LED = 2V (LED voltage)

I LED = 20 mA (current we want)

■ Example Calculation

R = (5V – 2V) / 0.02A
R = 3 / 0.02
R = 150 Ohms

■ We need a 150 Ohm resistor!

How to Connect LED & Resistor to Arduino

- Connect the Arduino pin to the resistor
- Connect the resistor to the LED long leg (+)
- Connect the LED short leg (–) to GND
- Double-check before powering on!

Simple Circuit Diagram

Arduino Pin 13 → Resistor → LED → GND

Electricity flows from the Arduino, through the resistor, lights the LED, and returns to ground.

■ Quiz Time!

Circle the letter of the best answer for each question.

1. What is Arduino?

A. A tiny computer that helps control lights, motors, and more
B. A type of resistor
C. A kind of fruit
D. A video game console

2. What does Arduino let us make?

A. Only boring things
B. Cool projects with electronics and coding
C. Sandwiches
D. Paper airplanes

3. What does LED stand for?

A. Light Emitting Diode
B. Light Energy Device
C. Long Electric Diode
D. Light Electric Detector

4. An LED glows when...

A. It gets cold
B. Electricity passes through it
C. It gets wet
D. It is shaken

5. Which of these is NOT a reason to use external LEDs?

A. Brighter light
B. Different colors available
C. Can be placed anywhere in a project
D. They never need any power

6. What does a resistor do in a circuit?

A. Makes the LED brighter forever
B. Controls the amount of electricity flowing
C. Stores electricity forever
D. Creates new electricity

7. What can happen to an LED WITHOUT a resistor?

A. It becomes cooler
B. It works perfectly forever
C. It might burn out
D. Nothing happens at all

8. What happens to an LED WITH a resistor?

A. It never lights up
B. It stays safe and glows just right
C. The Arduino stops working
D. It turns off immediately

9. Which formula is used to calculate resistor value?

A. R = V source × V LED
B. R = (V source – V LED) / I LED
C. R = I LED / V source
D. R = V source + I LED

10. In the example, what is V source (the Arduino voltage)?

A. 2V
B. 5V
C. 9V
D. 12V

11. In the example, what is V LED (the LED voltage)?

A. 5V
B. 3V
C. 2V
D. 1V

12. In the example, what is I LED (the current)?

A. 20 mA
B. 50 mA
C. 2 A
D. 100 mA

13. What resistor value did we calculate in the example?

A. 100 Ohms
B. 150 Ohms
C. 200 Ohms
D. 50 Ohms

14. Which leg of the LED is the long leg (+)?

A. The positive leg
B. The negative leg
C. The neutral leg
D. The ground leg

15. Which leg of the LED connects to GND?

A. The long leg
B. The short leg
C. The middle leg
D. Both legs at once

16. What should you always do before powering on your circuit?

A. Turn off all the lights
B. Double-check your connections
C. Remove the resistor
D. Disconnect the Arduino

17. In the simple circuit diagram, what comes right after Arduino Pin 13?

A. The LED
B. GND
C. The resistor
D. A battery

18. In the circuit, what happens after electricity passes through the resistor?

A. It returns to the store
B. It lights the LED
C. It stops flowing
D. It turns into water

19. Where does electricity go after it lights the LED?

A. Back to Pin 13
B. Into the resistor again
C. To ground (GND)
D. Nowhere — it disappears

20. Why do we need resistors in an LED circuit?

A. To make the circuit look nicer only
B. To keep the LED safe from too much electricity
C. To make the Arduino turn on
D. To change the LED's color

■ Answer Key

1. A	2. B
3. A	4. B
5. D	6. B
7. C	8. B
9. B	10. B
11. C	12. A
13. B	14. A
15. B	16. B
17. C	18. B
19. C	20. B

■ Great job learning about Arduino, LEDs, and resistors!

Keep Learning & Keep Building!